Lab 08: ILLUMINATION!

ECE 484: Embedded Systems Design

Grade	Score	Signature
Attendance: Gabe Litteken		
Attendance:		
Attendance:		
Demonstration	10	

Overview

In this lab, you will design an embedded system that will use an analog voltage to control the brightness level of an LED. The analog voltage will be generated by a voltage divider using a potentiometer. For debugging purposes, the current digital voltage will be output to an LCD module controlled by the I2C/TWI module. Along with the LCD module, three LEDs will be lit when the analog voltage exceeds particular thresholds.

Deliverables

1. System Demonstration (10 points)
2. Completed Worksheet submitted to Canvas (5 points)
3. Commented Source Code submitted to Canvas (5 points)

Instructions

Your embedded system will consist of the following hardware components:

1. An LCD screen connected to a I2C/TWI module. (Remember to use pull-up resistors for SCL and SDA lines).
2. Four LEDs with 330Ω current-limiting resistors. Three of these LEDs will be used to show that the voltage has exceeded particular thresholds while the other LED will have variable brightness controlled by an analog voltage.
3. A 10kΩ potentiometer with the tap of the potentiometer serving as the analog voltage for the system. The tap will be connected to an analog GPIO on your microcontroller and the anode of the brightness controlled LED.

Starter code for this lab is provided in the assignment page on Canvas. For the software, you will need to configure the ADC module to trigger an interrupt whenever the ADC has completed a conversion of your chosen analog channel. Remember that you will have to choose which analog channel your A/D module is converting; set the proper conversion rate for the A/D module; and choose whether your output data is left or right adjusted.

Once you have configured your A/D module, you will need to write the A/D conversion complete ISR. Within this ISR, you will need to read all 10 bits of ADCH and ADCL and store them in a variable. You will then write the 10-bit representation, in ***decimal form***, to the LCD such that the LCD reads "VD:ABCD" where ABCD is the quantized voltage reading from the potentiometer. After sending the quantized voltage to the LCD, you will then check whether the quantized voltage has exceeded 1.67V, 3.33V, and 4.8V. You will need to calculate the quantized values for each of these voltage levels and record them in the first question at the end of this manual. If the quantized voltage is below 1.67V, all three LEDs will be unlit; if the quantized voltage is between 1.67V and 3.33V, the first LED will be lit; if the quantized voltage is between 3.33V and 4.8V, the first and second LED will be lit; and if the quantized voltage is at least 4.8V, all three LEDs will be lit. Finally, after checking the quantized voltage levels, your A/D conversion complete ISR will trigger another conversion.

Demonstration Rubric

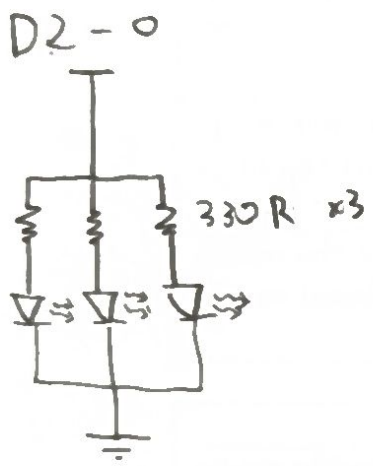
Please demonstrate your system to Dr. Swanson or the TA and they will record your demonstration score at the top of your lab manual. Checkpoint scores are provided in the following rubric. Scores between checkpoints may be awarded if a student has exceeded the previous checkpoint but has not reached the next checkpoint(s).

Score	Reasoning
10.0	Student can continuously convert analog voltage that changes when the dial of the potentiometer is rotated. The quantized voltage is correctly displayed on the LCD screen. The three quantization threshold LEDs are lit at the proper thresholds.
8.0	Student can continuously convert analog voltage that changes when the dial of the potentiometer is rotated. The quantized voltage is correctly displayed on the LCD screen.
6.0	Student can only convert analog voltage upon system startup. Rotating the potentiometer dial has no effect on the system. The quantized voltage is displayed on the LCD screen but the quantized voltage number is not correct.
4.0	Student has properly configured the A/D module.
2.0	Student has properly wired all necessary circuitry.
0.0	Student cannot program their microcontroller and has not installed microchip studio or has functional code in their `while(1)` loop.

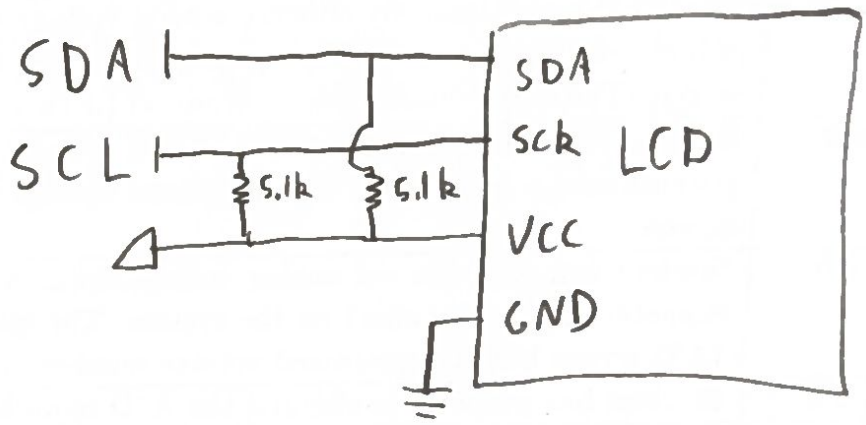
Circuit Diagram

Draw a circuit diagram for your system. Make sure to label all major modules, resistance values, and signal lines.

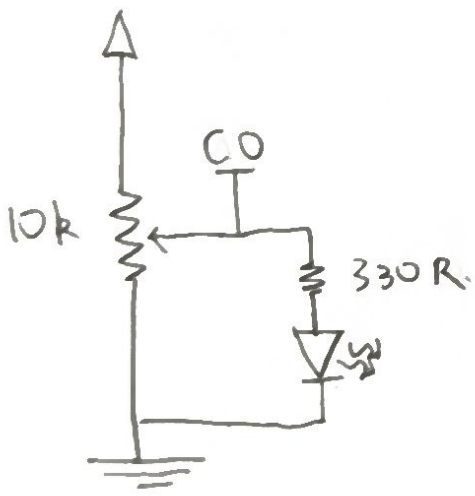
LEDs



LCD



Potentiometer



Design Questions

1. Record your calculations for the quantized voltage thresholds for the following analog voltages:

a.) 1.67V

$\frac{5.0}{1024} = \boxed{4.88 \text{ mV step size}}$

$\frac{1.67 \text{ V}}{4.88 \text{ mV}} = \boxed{342}$

b.) 3.33V

$\frac{3.33 \text{ V}}{4.88 \text{ mV}} = \boxed{682}$

c.) 4.8V

$\frac{4.8 \text{ V}}{4.88 \text{ mV}} = \boxed{964}$

2. Suppose we designed a system where we have analog inputs on analog channel 0 and channel 1 (PINC0:1). However, we can only read one analog channel per conversion. Write an A/D conversion complete ISR that is able to read both channels.

```
Global variable
bool ADCSelect = false;
```

```
ISR (ADC_vect) {
    uint16_t ADCVal = ADCL;
    ADCVal = ADCH << 8;
        ⋮ (other code)
    if (ADCSelect) {
        ADMUX |= (1<<MUX0);
    } else {
        ADMUX &= ~(1<<MUX0);
    }
    ADCSelect = !ADCSelect;
    ADCSRA |= (1<<ADSC);
}
```

3. Suppose we have a system where we need to read analog inputs on all six analog channels (PINC0:5). What effect would this have on the sampling rate of an individual analog channel?

The sampling rate of each channel would be divided by 6.

4. Why doesn't the A/D quantized voltage always reach 0 or 1023? What are the corresponding analog voltage values of the highest and lowest quantized voltages you observed?

There is always a small amount of resistance in wires and in the potentiometer. In addition, the LED connected to the potentiometer modifies the amount of voltage drop.

ADC min: 27

ADC max: 957